

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Mathematics
2. **Subject Name:** Optimization Techniques **L-T-P:** 3-0-0
Credits:

Commented [U1]: Please enter the name in Arial 12 font

Commented [U2]: Please enter the details (use Arial 12 font)

3. **Pre-requisites:** Void

Commented [U3]: Please enter the pre-requisites (subject name and subject no. in Arial 12 font)

4. **Syllabus and reference books:**

Commented [U4]: 1. Please type the syllabus and reference books in the box provided below.

Syllabus: Baseline models, linear programming problems, convex sets, convex functions and their properties ,
Basic feasible solution, optimal solution, related theorems, Graphical method for solving two and three variable problems ,
Simplex method ,
Artificial variable technique(Big-M method/Two phase method),
Duality theorems, Primal-dual simplex algorithm
Application in transportation and assignment problem
Classical optimization theory(Role of convexity in optimization theory, First and second order properties of convex function, First and second order optimality conditions for unconstrained and constrained optimization problem, KKT optimality condition, least mean square problem.

Please use "Arial" font with a font size of "12" and a line spacing of 1.0 for syllabus and reference books. Please use Justified Text (Ctrl+J) for the content entered in the boxes
Reference books should be numbered as 1), 2), 3)
Please provide all bibliographic details for the reference books

Reference Books:

References:

1. Taha, H. A. (2011). Operations research: an introduction (Vol. 790). Upper Saddle River, NJ, USA: Pearson/Prentice Hall.
2. F. Hillier and G. Lieberman (2005) Introduction to Operations Research, 8th edition, McGraw-Hill.
3. Linear and nonlinear programming by David Luenberger

Teachers: Prof A Goswami, Prof G Panda, Prof D Chakraborty, Prof S Khare

5. **Lecture-wise break-up:**

Commented [U5]: Please provide lecture-wise break-up in the table given below:

Please use "Arial" font with a font size of "12" and a line spacing of 1.0

Commented [U6]:

Sl. No.	Topic	No. of lectures
1.	Baseline models, linear programming problems, convex sets, convex functions and their properties . Basic feasible solution, optimal solution, related theorems, Graphical method for solving two and three variable problems [4 lectures]	4

2.	Simplex method	6
3.	Artificial variable technique(Big-M method/Two phase method)	4
4.	Duality theorems, Primal-dual simplex algorithm	4
5.	Application in transportation and assignment problem	4
6.	Classical optimization theory, Role of convexity in optimization theory	4
7.	First and second order properties of convex function, First and second order optimality conditions for unconstrained optimization problem	6
8.	Optimality conditions for constrained optimization problem, KKT optimality condition, least mean square problem	8
Total number of hours		40

Commented [U7]: Please convert the template into PDF and upload it in ERP